

# Expected Insurance Coverage and Pharmaceutical Innovation: Evidence from China's National Drug Price Negotiation Policy

Chenyuan Liu<sup>1</sup>, Yi Lu<sup>1</sup>, Ronghe Sun<sup>1</sup>, and Wanyu Yang<sup>2</sup>

<sup>1</sup>Tsinghua University and <sup>2</sup>Dongbei University of Finance and Economics

Discussant: Lucy Xiaolu Wang

University of Massachusetts Amherst; MPI - innovation; CCHE

# Main message and contribution

---

- Research question:  
How does combining drug price negotiation with expanded insurance coverage affect (Chinese) pharmaceutical innovation?
- Approach: stacked DID on drug sales; single-event TWFE DID on clinical trials (drugs vs. vaccines) at the disease level and firm level
  - Chinese clinical trials+ hospital sales (500+ hosp., 20 cities), 2013-2023
- Results: price negotiation + insurance coverage promotes innovation
  - Prices drop ~50%, revenues rise 187%; clinical trials increase by 0.56/disease/yr, driven by novel Type I drugs; small firm entry & collaboration increase
- Contribution: a systematic evaluation of the China NDPN policy and finds that price negotiation + market expansion promote innovation

# 1. Identification: control, obs. unit, decomp.

---

- Q1: How comparable are vaccines to drugs and therapeutic biologics?
  - Vaccines are excluded from NRDL by law, but also differ in overall incentives, market dynamics (routine immunization vs. therapeutic), and other policies (bioequivalent evaluation, central procurement) that affected mainly small molecule drugs
  - **Ideas:** (1) comparison across drug timing groups, or construct disease-exposure-rel. measures; (2) need much richer organizational and portfolio controls for firm analysis
- Q2: Observational unit and innovation analysis specification
  - 6,962 diseases at ICD-10 3-char. level -> extreme sparsity (mean 0.18); innovation uses TWFE with asymmetric window (3 pre, 2 post), not stacked DID like revenue
  - **Ideas:** (1) coarser disease level and extend post-window (Miller 2023 JEP: min. 3 pre & post); (2) pre-trend tests (some underpowered) and robustness w not-yet-treated as control; (3) revenue analysis: optimal reweighting for stacked DID (Wing et al. 2024)
- Q3: How to disentangle price negotiation from insurance coverage?
  - DID est. combined effect of price cuts and NRDL inclusion, which matters more?
  - **Ideas:** (1) exploit provincial coinsurance rate variation; (2) compare diseases with different NRDL exposure timing

## 2. Measures, classification, & firm-level analysis

---

- Q1: Why not try ATC-year-level analysis instead of disease-year-level?
  - Paper uses ATC-4 codes to remove substitutes (p.7, Dubois & Lasio 2018) but claims insufficient ATC for main analysis (p.11); disease level has m-n mapping problems (drug can have many indications, off-label use, *vice versa*)
  - **Ideas:** (1) report ATC missing rate and whether missing is systematic, and subsample robustness check using trials with non-missing ATC codes; (2) construct some ATC counterpart (or reconstruction based on clinical info)
- Q2: Firm classification & firm-level analysis needs deepening
  - “Disease-by-group” as “research field” isn’t standard in the lit.; collaboration defn worth refining; firm type (CDMO/CRO vs. originator) may contaminate domestic innovation narrative; no portfolio controls (patents, exp., scope)
  - **Ideas:** (1) define research fields by distinct ATC codes in a trial (can vary by phase); (2) separate CDMO/CRO from originators, more firm-level controls; (3) # unique ATCs can be used for a pivot analysis (e.g., Jungbauer et al. 2026)

# Minor suggestions

---

- New entry result is cool but also feels “too good” given the fixed costs
  - Does the result purely reflect a natural pull with expanded market size, or also w push factors?
- More results using Poisson specification or a stacked Poisson analysis
  - Existing 3 Poisson estimates aren’t very consistent with main results (Type I est. no longer sig.)
- First-in-China vs. first-in-world distinction matters for innovation interpretation
  - This analysis can complement that for Type I drugs and firm origin analysis
- Welfare analysis: incorporate government expenditure in back-of-envelope calc.?
  - Potential future project: what’s the further downstream health impact? Fiscally sustainable?
- Check accuracy of cited literature framing
  - E.g., Agha et al. (2022) on insurance design; Dix & Lensman (2025) on cancer drug cocktails
- Appendix could be condensed with more compact and salient comparison
  - Currently have many figures and tables that can be combined to improve readability/clarity

# Final thoughts

---

- Very interesting and very timely topic
- Key takeaway: drug price negotiation paired with expanded insurance coverage can promote innovation when market expansion more than compensates for price reductions
- Main thoughts: strengthen identification (new control group, disentangle channels); consider ATC-level analysis as robustness; deepen firm-level analysis with portfolio controls
- Looking forward to a newer version!